
Report 1

Simulation of diffusion processes

Toshihiro Yamada
Leto Van Ruymbeke

Framework

We work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, $(W_t)_t$ is supposed to be a standard Brownian motion of filtration $(\mathcal{F}_t)_t$.

We consider a diffusion process $(X_t)_t$ that verifies :

$$(\text{SDE}) \quad \begin{cases} dX_t = b(X_t)dt + \sigma(X_t)dW_t \\ X_0 = x \end{cases}$$

We want to compute $\mathbb{E}^{\mathbb{P}}[f((X_t)_{t \in [0, T]})]$ for a given function f .

b and σ are supposed to be “appropriate” : the exact conditions they have to respect are determined in the **Sufficient Conditions part**.

1. Equivalent equation, constant volatility

$(X_t)_t$ is the solution of (SDE), where we assume b and σ respect the sufficient conditions for that solution to **exist** and be **unique**. Under those, X_t is a Itô process. We define $Y_t = g(X_t)$, where $g(x) = \int^x \frac{1}{\sigma(y)} dy$.

We also assume $g \in C^2$. By Itô formula:

$$dY_t = g'(X_t)dX_t + \frac{1}{2}g''(X_t)\sigma^2(X_t)dt$$

where $g'(x) = \frac{1}{\sigma(x)}$ and $g''(x) = -\frac{\sigma'(x)}{\sigma^2(x)}$.

It yields :

$$\begin{aligned} dY_t &= \frac{1}{\sigma(X_t)}(b(X_t)dt + \sigma(X_t)dW_t) - \frac{1}{2} \frac{\sigma'(X_t)}{\sigma^2(X_t)} \sigma^2(X_t)dt \\ &= \left(\frac{b(X_t)}{\sigma(X_t)} - \frac{1}{2} \sigma'(X_t) \right) dt + dW_t \end{aligned}$$

We deduce :

$$dY_t = \left[\frac{b(g^{-1}(Y_t))}{\sigma(g^{-1}(Y_t))} - \frac{1}{2} \sigma'(g^{-1}(Y_t)) \right] dt + dW_t$$

from which we define $a(y) = \frac{b(g^{-1}(y))}{\sigma(g^{-1}(y))} - \frac{1}{2} \sigma'(g^{-1}(y))$

Sufficient conditions

Let's now define what we mean by "appropriate" for b and σ :

- **Existence of X_t** - there exists K such that, $\forall x, y$:
 - *Lipschitz* : $|b(x) - b(y)| + |\sigma(x) - \sigma(y)| \leq K|x - y|$
 - *Linear growth* : $|b(x)| + |\sigma(x)| \leq K(1 + |x|)$
- $g \in C^2$: We need σ' to be continuous and σ^2 not vanishing. We take the sufficient condition : $\begin{cases} \sigma \in C^1 \\ \forall x, \sigma(x) \neq 0 \end{cases}$
- g^{-1} **existence** : g must be a bijection on its domain. We thus need g to be monotone $\iff g' > 0$ (or $g' < 0$) $\iff \sigma > 0$ (or $\sigma < 0$), which is always the case since we supposed that σ is continuous **and** doesn't vanish.

Conclusion : next we keep the following conditions as verified by b and σ :

- b K -lipschitz, linear growth of weight K
- $\sigma \in C^1$, $\sigma > 0$, K -lipschitz, linear growth of weight K

They allow Y_t to exist.

2. Girsanov : drift removal by change of probability

In order to apply *Girsanov-Maruyama*, we need :

$$\int_0^T a(Y_t)^2 dt < \infty \iff t \mapsto \frac{b(g^{-1}(Y_t))}{\sigma(g^{-1}(Y_t))} - \frac{1}{2}\sigma'(g^{-1}(Y_t)) \in L^2([0, T])$$

$$1. \iff \frac{b}{\sigma} - \frac{1}{2}\sigma' \in L^2\left(Dom_\omega(X)\right)$$

where $Dom_\omega(X) = \{X_t(\omega), t \in [0, T]\}$ is the domain X takes its values in, which still needs to be determined. We usually ask $\sigma \in C_b^1$, i.e. σ and σ' bounded, to ensure :

$$\int_0^T a(Y_t)^2 dt < \infty$$

2. $L_t = \exp\left(-\int_0^t a(Y_s)dW_s - \frac{1}{2}\int_0^t a(Y_s)^2 ds\right)$ to be a $\mathbb{P}$ -martingale.

Here is the proof that $(L_t)_t$ is indeed a martingale, and under which conditions.

Let's define $Z_t = -\int_0^t a(Y_s)dW_s - \frac{1}{2}\int_0^t a(Y_s)^2 ds$. We want to compute dL_t .

As $L_t = \exp(Z_t)$, by Itô formula :

$$\begin{aligned}
dL_t &= \exp(Z_t)dZ_t + \frac{1}{2}\exp(Z_t)d\langle Z, Z \rangle_t \\
&= L_t \left(-a(Y_t)dW_t - \frac{1}{2}a^2(Y_t)dt \right) + \frac{1}{2}L_t a^2(Y_t)dt \\
&= -a(Y_t)L_t dW_t
\end{aligned}$$

$$\text{So } L_t = L_0 + \int_0^t (-a(Y_s)L_s)dW_s = 1 + \int_0^t (-a(Y_s)L_s)dW_s$$

So L_t is a local martingale, and with the next additional condition that we consider to be verified, L_t is a $\mathbb{P}$ -martingale :

$$\mathbb{E}^{\mathbb{P}} \left[\exp \left(\frac{1}{2} \int_0^T a^2(Y_s)ds \right) \right] < +\infty$$

In the end, assuming that last condition also ensures that $\int_0^T a(Y_t)^2 dt < \infty$, so we take it for both *Girsanov-Maruyama* hypotheses to be verified. According to this theorem, under $\mathbb{Q}$ of density L_T with respect to $\mathbb{P}$:

$$W_t^{\mathbb{Q}} = W_t + \int_0^t a(Y_s)ds \text{ is a standard Brownian motion}$$

But as we know, $\mathbb{E}^{\mathbb{P}}[f((X))] = \mathbb{E}^{\mathbb{P}}[f(g^{-1}(Y))]$ and $\frac{d\mathbb{Q}}{d\mathbb{P}} = L_T$, which means :

$$\begin{aligned}
\forall A \in \mathcal{F}, \quad \mathbb{Q}(A) &= \int_A L_T d\mathbb{P}(\omega) \text{ by definition of } L_T \\
&= \mathbb{E}^{\mathbb{P}}[L_T \mathbf{1}_A]
\end{aligned}$$

Thus, for all random variable $Z : \Omega \rightarrow \mathbb{R}$,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}[Z] &= \int_{\Omega} Z(\omega) d\mathbb{Q}(\omega) \\
&= \int_{\Omega} Z(\omega) L_T d\mathbb{P}(\omega) \\
&= \mathbb{E}^{\mathbb{P}}[L_T Z]
\end{aligned}$$

and if $L_T > 0$, which is the case here : $\mathbb{E}^{\mathbb{P}}[Z] = \mathbb{E}^{\mathbb{Q}}\left[\frac{Z}{L_T}\right]$.

Now we take $Z := f(g^{-1}(Y))$:

$$\begin{aligned}\mathbb{E}^{\mathbb{P}}[f(X)] &= \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y)) \cdot e^{\int_0^T a(Y_s) dW_s + \frac{1}{2} \int_0^T a^2(Y_s) ds} \right] \\ &= \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y)) \cdot e^{\int_0^T a(Y_s) dW_s^{\mathbb{Q}} - \frac{1}{2} \int_0^T a^2(Y_s) ds} \right]\end{aligned}$$

$$\text{since } dW_t^{\mathbb{Q}} = dW_t + a(Y_t)dt$$

Note : the paper given by Yamada-sensei seems to have used dW_t when $dW_t^{\mathbb{Q}}$ is intended for this last result. But the measure change $dW_t \rightarrow dW_t^{\mathbb{Q}}$ requires a $-\frac{1}{2}$ instead of a $\frac{1}{2}$. The final equation being :

$$\mathbb{E}^{\mathbb{P}}[f(X)] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y)) \cdot e^{\int_0^T a(Y_s) dW_s^{\mathbb{Q}} - \frac{1}{2} \int_0^T a^2(Y_s) ds} \right]$$

Sufficient conditions update

We add to the Sufficient Conditions found before :

- **Girsanov-Maruyama conditions** : $\begin{cases} \sigma' \text{ and } b \text{ bounded} \\ \exists \alpha, \sigma \geq \alpha > 0 \end{cases}$

Note : Those conditions ensures $\mathbb{E}^{\mathbb{P}} \left[\exp \left(\frac{1}{2} \int_0^T a^2(Y_s) ds \right) \right] < +\infty$, which is sufficient to apply Girsanov-Maruyama. Weaker conditions with the same consequences can be found, but we keep those as they are both simple and apply to standard models, like Black-Scholes.

3. Simplifying implementation : $dW^{\mathbb{Q}}$ integral removal

The presence of the integral with respect to $dW_t^{\mathbb{Q}}$ is a hindrance to computation. In practice, deterministic integrals are easy to approach but stochastic ones need us to discretize Brownian motions and approximate Itô integrals. Thus we suppose a has a primitive h .

We define $Z_t = h(Y_t)$. By Itô formula :

$$dZ_t = h'(Y_t) dY_t + \frac{1}{2} h''(Y_t) d\langle Y, Y \rangle_t$$

$$\begin{aligned}
\text{while } dY_t &= a(Y_t)dt + dW_t \\
&= a(Y_t)dt + dW_t^{\mathbb{Q}} - a(Y_t)dt \\
&= dW_t^{\mathbb{Q}}
\end{aligned}$$

Thus, $(Y_t)_t$ is a Brownian motion under $\mathbb{Q}$, and $d\langle Y, Y \rangle_t = dt$, so :

$$\begin{aligned}
dZ_t &= a(Y_t)dW_t^{\mathbb{Q}} + \frac{1}{2}a'(Y_t)dt \\
\Rightarrow Z_T - Z_0 &= \int_0^T a(Y_s)dW_s^{\mathbb{Q}} + \frac{1}{2} \int_0^T a'(Y_s)ds \\
\Rightarrow h(Y_T) - h(Y_0) &= \int_0^T a(Y_s)dW_s^{\mathbb{Q}} + \frac{1}{2} \int_0^T a'(Y_s)ds \\
\Rightarrow \int_0^T a(Y_s)dW_s^{\mathbb{Q}} &= h(Y_T) - h(Y_0) - \frac{1}{2} \int_0^T a'(Y_s)ds
\end{aligned}$$

In the end, we have :

$$\mathbb{E}^{\mathbb{P}}[f(X)] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y)) \cdot e^{h(Y_T) - h(Y_0) + \int_0^T -\frac{1}{2}(a'(Y_s) + a(Y_s)^2)ds} \right]$$

from which we define $c := -\frac{1}{2}(a' + a^2)$:

$$\mathbb{E}^{\mathbb{P}}[f(X)] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y)) \cdot e^{h(Y_T) - h(Y_0) + \int_0^T c(Y_s)ds} \right]$$

4. Code

4. 1. Monte-Carlo under Euler scheme

4. 1. 1. Theory

First we implement the function giving the exact Black-Scholes price for a European call, in *black_sholes.py*.

In *models.py*, we define a *Python* Class named *BSmodel* which represents the classic Black-Scholes class EDS with :

- S_0 , the starting point
- $b : x \mapsto rx$ the *drift*
- $\sigma : x \mapsto \sigma x$ the *volatility*

Then, in *monte_carlo.py*, we define a first Monte-Carlo function that, given a model, a maturity, a payoff function, a number of steps and a number of paths, discretizes the model using Euler scheme to estimate $E[f(X)]$ and $\text{Var}(f(X))$ with the usual estimators. The induction gives :

$$\begin{aligned}\hat{X}_{k+1} &= \hat{X}_k + b(\hat{X}_k)\Delta t + \sigma(\hat{X}_k)\sqrt{\Delta t}Z_k \quad (*) \\ &= \tilde{X}_k(1 + r\Delta t + \sigma\sqrt{\Delta t}Z_k) \text{ for B-S solution}\end{aligned}$$

where $\Delta t = \frac{T}{N}$ and $Z_k \sim \mathcal{N}(0, 1)$

A few things to note:

- This method gives us $\hat{\mu}_M = \frac{1}{M} \sum_{j=1}^M f(\hat{X}_j)$
- The variance of this method is $\hat{\sigma}_M^2 = \text{Var}(\hat{\mu}_M) = \sigma_{\text{Euler}}^2/M$, where $\sigma_{\text{Euler}} = \text{Var}(f(\hat{X}))$.
- We have the exact induction equation : $X_{k+1} = X_k e^{(r - \frac{\sigma^2}{2})\Delta t + \sigma Z_k}$, because the solution of B-S is just $X_t = S_0 e^{(r - \frac{\sigma^2}{2})\Delta t + \sigma W_t}$. Thus, (*) is just a Taylor expansion at $O(\Delta t)$ ($= O(\frac{1}{N})$).
- The exact solution and the approximation differ from Taylor expansion at order Δt^2 and higher. r increases that difference. Indeed :

$$\begin{aligned}\mathbb{E}[X_{k+1}|X_k] &= X_k \cdot \mathbb{E}\left[\exp\left(\left(r - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}Z_k\right)\right] \\ &= X_k \exp\left(\left(r - \frac{1}{2}\sigma^2\right)\Delta t\right) \cdot \mathbb{E}\left[e^{\sigma\sqrt{\Delta t}Z_k}\right] \\ &= X_k e^{(r - \frac{1}{2}\sigma^2)\Delta t} e^{\frac{1}{2}\sigma^2\Delta t} \\ &= X_k e^{r\Delta t}\end{aligned}$$

$$\mathbb{E}[\hat{X}_{k+1}|\hat{X}_k] = \hat{X}_k(1 + r\Delta t)$$

$$\Rightarrow \mathbb{E}[X_{k+1}|X_k] - \mathbb{E}[\hat{X}_{k+1}|\hat{X}_k] = \frac{r^2}{2}X_k\Delta t^2 + O(\Delta t^3)$$

- Finally, as for any $p > 0$, $O(\Delta t^p) = O(\frac{1}{N^p})$, there exists a **bias** between the exact and the Euler solution that decreases when the grid becomes denser.

4. 1. 2. First tests' results

In *main.py*, we started by implementing a function named *validate*. It takes a set of parameters m, f, T, N, M and *ref* (see the code for more details), computes both $\hat{\mu}_M$ and $\hat{\sigma}_M$ defined earlier. Finally, *validate* gives true if the validating condition is verified :

$$|\mu - \hat{\mu}_M| \leq 4\hat{\sigma}_M$$

We define n_σ by :

$$n_\sigma = \frac{|\mu - \hat{\mu}_M|}{\hat{\sigma}_M}$$

We can show that there is a 0.0063% chance for a given instance not to pass this test. The reason why we need it to be so low is that the Monte-Carlo method will run hundreds of these tests for different values of the parameters, so the probability of a failure will increase. Thus **we take $4\hat{\sigma}$ as a threshold** and not $3\hat{\sigma}$ or $2\hat{\sigma}$, that would be satisfied with high probability for one test, but not for hundreds of them.

Then we run this test for a decent number of instances. As such, we executed *testMC()*, that runs *validate(m,f,T,N,M,ref)* for every possible combination of the parameters given by the next chart :

Parameter	Tested values
T	1
S_0	100
r	0.01, 0.05, 0.1, 0.4
σ	0.1, 0.2, 0.3, 0.5
K	$S_0, S_0 - 20, S_0 + 20$
N	10, 100, 1000
M	$10, 10^2, 10^3, 10^4, 10^5$

Table 1: **Initial test – Monte-Carlo coarse parameters for Euler scheme**

testMC gives back nothing, but prints every failure and for what specific set of parameters it occurred. This allows us to study if these failures were statistically expected, or if there is a bug in our code.

Before presenting the results, one important thing to mention is that the test held here has false-negative outcomes. Indeed, what we want is for this test to warn us when our estimation, $\hat{\mu}_M$, is too far from μ that we try to estimate. However,

for big values of M (large amount of samples $X_T^{(i)}$), $\hat{\sigma}_M = \hat{\sigma}/\sqrt{M}$ decreases so the test window becomes more and more narrow. On the other hand, increasing M makes $\hat{\mu}_M$ closer to $\mu_{\text{Euler}} = \mathbb{E}[f(\hat{X})] \neq \mu$. This means that, as long as the bias $|\mu_{\text{Euler}} - \mu|$ is low, meaning the grid is thin (big N), the strictness of this test increasing with M bears no issue. If the test has too many failures, there is high probability that there is an error in our code.

But, if the grid is too coarse (low N), the bias $|\mu_{\text{Euler}} - \mu|$ increases. Then, increasing M makes our estimator $\hat{\mu}_M$ closer to μ_{Euler} (not of μ), but also makes our test of $|\hat{\mu}_M - \mu|$ more severe. In this specific case, we can't ensure our test refused an instance because the code is wrong, or because M is too big compared to N .

This has to be considered when reading the test failures.

	Parameters					True Price	M-C results		Test results	
Error type	r	volatility	K	N	M	μ	$\hat{\mu}_M$	$\hat{\sigma}_M$	$ \hat{\mu}_M - \mu $ (\$-unit)	$ \hat{\mu}_M - \mu $ ($\hat{\sigma}_M$ -unit)
Type 1	0.4	0.1	100	10	10^4	49.1826	47.9543	0.1434	1.23	8.57σ
	0.4	0.1	100	10	10^5	49.1826	48.0467	0.0452	1.14	25.13σ
	0.4	0.1	80	10	10^4	69.1825	68.1714	0.1417	1.01	7.14σ
	0.4	0.1	80	10	10^5	69.1825	68.0241	0.0452	1.16	25.63σ
	0.4	0.1	120	10	10^4	29.2522	28.1350	0.1401	1.12	7.97σ
	0.4	0.1	120	10	10^5	29.2522	28.0998	0.0445	1.15	25.90σ
	0.4	0.2	100	10	10^5	49.3891	48.0755	0.0895	1.31	14.68σ
	0.4	0.2	80	10	10^4	69.1880	67.9651	0.2850	1.22	4.29σ
	0.4	0.2	80	10	10^5	69.1880	68.1753	0.0907	1.01	11.17σ
	0.4	0.2	120	10	10^5	31.0551	29.8818	0.0826	1.17	14.20σ
	0.4	0.3	100	10	10^5	50.7238	49.6940	0.1308	1.03	7.87σ
	0.4	0.3	80	10	10^4	69.4062	67.3628	0.4276	2.04	4.78σ
	0.4	0.3	80	10	10^5	69.4062	68.3606	0.1365	1.05	7.66σ
	0.4	0.3	120	10	10	34.6379	10.6880	5.1087	23.95	4.69σ
	0.4	0.3	120	10	10^4	34.6379	33.1092	0.3719	1.53	4.11σ
	0.4	0.3	120	10	10^5	34.6379	33.4654	0.1186	1.17	9.89σ
	0.4	0.5	100	10	10^5	56.3896	54.8896	0.2128	1.50	7.05σ
	0.4	0.5	100	10^3	10	56.3896	22.6136	5.5556	33.78	6.08σ
	0.4	0.5	120	10	10^4	43.5771	40.8770	0.6067	2.70	4.45σ
	0.4	0.5	120	10	10^5	43.5771	42.7537	0.1993	0.82	4.13σ
Type 2	0.01	0.1	120	10	10	0.1900	0	0	0.19	∞
	0.01	0.1	120	100	10	0.1900	0	0	0.19	∞
	0.05	0.2	120	10	10	3.4140	0	0	3.41	∞
	0.1	0.1	120	100	10	1.3276	0	0	1.33	∞
	0.1	0.2	120	100	10	5.2034	0	0	5.20	∞
Type 3	0.01	0.1	120	100	100	0.1900	0.0243	0.0242	0.17	6.85σ
	0.01	0.3	100	100	10	12.4926	1.3549	0.9215	11.14	12.09σ
	0.01	0.5	100	10^3	10	20.3469	2.6791	2.4375	17.67	7.25σ
	0.01	0.5	80	10^3	10	30.2268	4.1345	2.6078	26.09	10.01σ
	0.01	0.5	120	100	100	13.5696	5.8586	1.8098	7.71	4.26σ
	0.05	0.3	100	10^3	10	14.9609	0.3481	0.3302	14.61	44.25σ
	0.05	0.3	120	100	10	7.2580	0.5988	0.5366	6.66	12.41σ
	0.05	0.3	120	10^3	10	7.2580	1.7421	1.2793	5.52	4.31σ
	0.1	0.2	120	10^3	10	5.2034	0.9644	0.8515	4.24	4.98σ
	0.1	0.5	120	100	10	18.3132	3.8925	3.1662	14.42	4.55σ

Table 2: **Monte-Carlo test failures**
Failures obtained out of the 720 combination of parameters.

4. 1. 3. Discussion

For 720 tests, we got ~30 failures, while we expected 0 or 1. Let's analyse these errors.

- **Type 1** : $r = 0.4$

The most common failures happen when the value of r is much over the usual values in finance : r is the riskless asset rate, it is never worth this much. But as explained in the previous remark, there is also a mathematical reason why many failures occurred for such a high value of r : Euler scheme has a bias that increases with r . Thus, restricting to $r \leq 0.1$ for our tests is a mathematically justified decision that should avoid this type of failures, while being closer to the reality of the market.

- **Type 2** : $M = 10, K > S_0$ (OTM)

Too few samples for the Monte-Carlo bears the risk of a null standard deviation, thus leading to an infinite gap when expressed in $\hat{\sigma}_M$ -unity. The test fails, when in truth a case where $\hat{\sigma}_M = 0$ shouldn't even be considered. This happens only because we have a degenerate sample : K is too high, compared to S_0 , to ensure that, among the 10 paths (too few), at least one will survive. This case teaches us that a condition has to assert that the samples are various enough. We take $M \geq 100$ to diminish the probability of $\hat{\sigma}_M = 0$.

- **Type 3** : The remaining cases

1. Too low N for a too high M .

Among the remaining cases, the first two rows fall in **Type 1**. Indeed, even if r is not too high for those, what matters is that Type 1 failures are due to **Euler scheme bias**. This bias grows **both** with r and with $O(\frac{1}{N})$. Thus, the first two rows, which display high M values with low N values, suffered from $N = 10$ while the passing condition of the test was narrow because of M being high ($\hat{\sigma}_M \xrightarrow{M \rightarrow \infty} 0$). We will run additional test with $N \geq 100$, which should decrease the bias.

2. μ is still skewed to the right while M is too small.

For the last 6 rows of the chart, that we do not re-categorize as being of Type 1, here is what we observe : M is low (10 to 100) and $K > S_0$. We know that $f(S_T)$ is skewed to the right because of the structure of the payoff (European call). Most of its values are around 0^+ , but a few far

exceed this domain. Thus, while the median of the distribution is close to 0, $\mu = \mathbb{E}[f(S_T)]$ gets pulled to the right by the rare big values taken by the distribution. See the next graph for a better understanding.

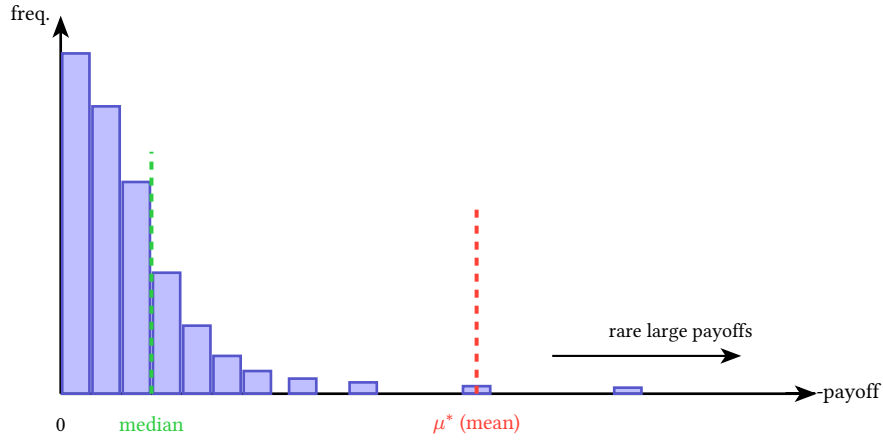


Figure 1: Payoff distribution — skewed by rare large outcomes.

The distribution is heavily concentrated near zero. A few rare but large payoffs pull the mean μ^ far to the right of the median.*

As we have a small M , we have a small sample, so a lower probability for the Euler distribution to **catch the tail** of huge values taken by the exact distribution. This leads to have $\mu > \hat{\mu}_M$ in almost all cases, because of the right-skew of the sampling distribution. Now, in this Type 3 case especially, $K > S_0$ (OTM), so the gap between μ and $\hat{\mu}_M$ widens with high probability: the numerator of n_σ becomes bigger, hence a lower probability to pass the test.

For these two reasons, the test becomes hard to pass under such conditions, independently of the correctness of the scheme. Again, increasing M seems to be the solution.

4. 1. 4. Second tests

The previous work taught us that :

- r must remain low, in accordance with both the reality of the market and to ensure the bias of Euler scheme doesn't prevent the results accuracy.
- M needs to be higher than 100 because it can lead to a null standard deviation which makes the tests fail in every case. Also, there is a risk that, for OTM cases, the Euler approximated variables $\hat{X}$ are too few to catch the high value taken rarely by the exact distribution, leading $\hat{\mu}_M$ to fail the test.
- N must be bigger than 100 to ascertain that Euler bias remain low.

So here are the new parameters sets :

Parameter	Tested values
T	1
S_0	100
r	0.01, 0.05, 0.1
σ	0.1, 0.2, 0.3, 0.5
K	$S_0, S_0 - 20, S_0 + 20$
N	100, 1000
M	$10^2, 10^3, 10^4, 10^5$

Table 3: **Second test – Monte-Carlo adjusted parameters for Euler scheme**

By running these tests a dozen of times, **no failure occurred**. The Euler scheme has been validated through suitable tests : it approximates μ in accordance with n_σ under a treshhold of 4.

Moreover, we can plot the bias of Euler scheme depending on N, under log-log scale, to confirm that our scheme bias converges in $O(\frac{1}{N})$. Here is what we get :

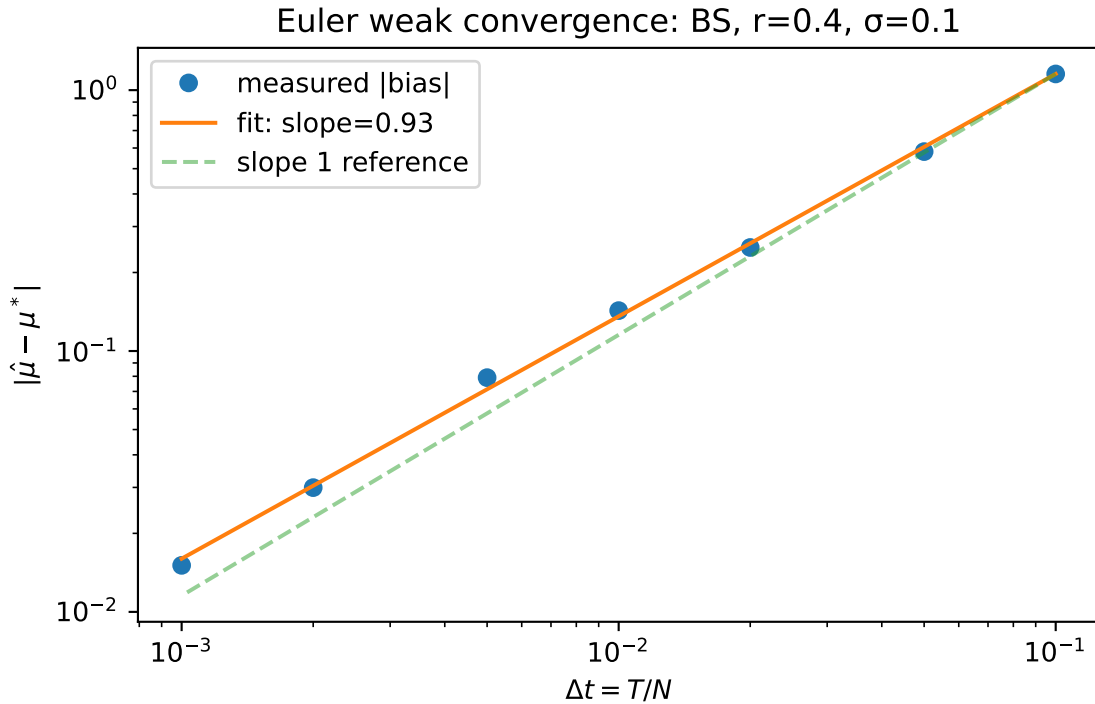


Figure 2: Euler convergence compared to $1/N$

We get a slope of 0.93 when we expected 1 : the bias empirical weak order is suitable compared to $1/N$.

4. 2. Beskos-Roberts

4. 2. 1. Implementation

The previous approximation method gives us a solid way to assert the next one's accuracy and allows us to detect errors in our code. The goal now is to use the final form of $\mathbb{E}^{\mathbb{P}}[f(X)]$ we got in Section 3. Let's remind it :

$$\mathbb{E}^{\mathbb{P}}[f(X)] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y)) \cdot e^{h(Y_T) - h(Y_0) + \int_0^T c(Y_s) ds} \right] \quad (*)$$

There is one main difficulty with the implementation of this formula, that is the $e^{\int c(Y)} : (c(Y_s))_{s \in [0, T]}$ is a random path, which means it needs to be simulated and approximated by a scheme that would let a new bias appear. Initially, the purpose of all those steps was to find a way to compute $E[f(X)]$ that wouldn't come with any bias. Let us define :

$$\Lambda := \int_0^T -c(Y_s) ds$$

As Λ depends on Y , it is random. But let's see it as a constant for a moment : the idea is that Λ could be the parameter of a **Poisson law**. We recall that :

$$\tilde{Z} \sim \mathcal{P}(\lambda) \iff \forall k \in \mathbb{N}, \mathbb{P}(\tilde{Z} = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Thus, if $Z \mid Y \sim \mathcal{P}(\Lambda)$, we have $\mathbb{P}(Z = 0 \mid Y) = \exp(-\Lambda) = e^{\int c(Y)}$, so we need to define $Z \mid Y$ as a Poisson law : the parameter Λ can depend on space and time, but once we fixed the path Y , Λ is fixed too.

Now, what experiment could be a Poisson process of parameter Λ ? We can build one using rejection sampling :

- Simulate an integer $P \sim \mathcal{P}(TB)$, where T is maturity and B is an upper bound of $-c$ (see the definition of c)
- Pick P coordinates on the time axis under $\mathcal{U}([0, T])$
- Pick P coordinates on the space axis under $\mathcal{U}([0, B])$
- That gives us a sequence of P points $(\tau_i, V_i)_{1 \leq i \leq P}$
- A point is kept when :

$$V_i < -c(Y_{\tau_i})$$

Let us introduce this new theorem :

Theorem : Thinning

Let there be a Poisson process of intensity $\lambda(\cdot)$ from which we independently keep each point with probability $p(\cdot)$. The surviving points form a Poisson process of intensity $\lambda(\cdot)p(\cdot)$

By doing this, conditionally to a path $(Y_s)_s$, the number of surviving points $Z \mid Y$ is a Poisson random variable of parameter Λ :

- Our test has a probability of success :

$$\mathbb{P}(\{\text{no points are under } -c(Y)\} \mid Y) = \mathbb{P}(Z = 0 \mid Y)$$

- For our case, $p := \int_0^T -c(Y_s)ds / TB$ (the area **under** the curve over the total area) and $\lambda := TB$ in the *thinning theorem*.

Hence, $\mathbb{P}(Z = 0 \mid Y) = e^{-\Lambda}$

Note : To get Y_{τ_i} we need a specific tool called the **Brownian bridge**. In our case, $Y_{\tau_i} \approx \mathcal{N}(Y_0, \tau_i)$, because it is not any normal distribution of variance τ_i : we have to take in consideration the fact that $\begin{cases} Y_0 = g(X_0) \\ Y_T \text{ is known} \end{cases}$.

Thus, we use the Brownian bridge. For a Brownian motion of maturity T , we have :

$$(W_t \mid W_0 = x, W_T = y) \sim \mathcal{N}\left((y - x)\frac{t}{T} + x, \frac{t(T - t)}{T}\right)$$

Eventually, our Monte-Carlo computes the wanted expectation by :

- Computing $f(g^{-1}(Y))$
- Simulating $Y_T \sim \mathcal{N}(0, T)$: this gives $e^{h(Y_T) - h(Y_0)}$
- Simulating $P \sim \mathcal{P}(TB)$ and P independent points (τ_i, V_i) with $\begin{cases} \tau_i \sim \mathcal{U}([0, T]) \\ V_i \sim \mathcal{U}([0, B]) \end{cases}$
- Simulating $(Y_{\tau_i} \mid Y_T, Y_0) \sim \mathcal{N}\left((y - x)\frac{t}{T} + x, \frac{t(T - t)}{T}\right)$
- Computing $\mathbb{1}_{\{Z=0\}} = \mathbb{1}_{\{\forall i, -c(Y_{\tau_i}) \leq V_i\}}$, which is exactly $\exp\left(\int_0^T c(Y_s)ds\right)$

- Multiply the three highlighted terms

By repeating the operation and taking the mean and the standard deviation, our Monte-Carlo becomes an unbiased estimator of $E[f(X)]$.

4. 2. 2. Results on Black-Scholes with European call

First, we tested our code on the Black-Scholes model under European call payoff, to compare Beskos-Roberts with the exact value, that we can compute. Here are the results.

Parameters				True Price	M-C results		Test results
r	volatility	K	M	μ	$\hat{\mu}_{BR}$	$\hat{\sigma}_{BR}$	Passed (P)/ Failed (F)
0.01	0.1	100	10^3	4.5303	4.5511	0.2256	P
			10^5		4.5412	0.0222	P
		120	10^3	0.1900	0.2072	0.0506	P
			10^5		0.1895	0.0044	P
	0.3	100	10^3	12.4926	12.4639	0.6016	P
			10^5		12.5213	0.0595	P
		120	10^3	5.7718	5.7494	0.4353	P
			10^5		5.8014	0.0423	P
	0.5	100	10^3	20.3469	20.3115	0.9988	P
			10^5		20.3827	0.0986	P
		120	10^3	13.5696	13.5266	0.8593	P
			10^5		13.6071	0.0842	P
0.05	0.1	100	10^3	7.1539	7.5519	0.5442	P
			10^5		7.1806	0.0486	P
		120	10^3	0.4862	0.6194	0.1889	P
			10^5		0.4845	0.0138	P
	0.3	100	10^3	14.9609	15.0453	0.7845	P
			10^5		14.9972	0.0764	P
		120	10^3	7.2580	7.3361	0.5877	P
			10^5		7.2918	0.0561	P
	0.5	100	10^3	22.9099	22.9451	1.1742	P
			10^5		22.9512	0.1148	P
		120	10^3	15.5413	15.5494	1.0242	P
			10^5		15.5851	0.0991	P
0.1	0.1	100	10^3	11.3923	11.5734	2.1042	P
			10^5		11.3635	0.1421	P
		120	10^3	1.3276	1.8545	1.0787	P
			10^5		1.2893	0.0529	P
	0.3	100	10^3	18.4941	18.5152	1.1120	P
			10^5		18.5441	0.1064	P
		120	10^3	9.5114	9.5763	0.8648	P
			10^5		9.5528	0.0807	P
	0.5	100	10^3	26.4431	26.6748	1.4475	P
			10^5		26.5083	0.1399	P
		120	10^3	18.3132	18.4758	1.2794	P
			10^5		18.3765	0.1223	P

Table 4: **Monte-Carlo test failures for Beskos-Roberts**

Out of the 36 possible combinations of parameters, no failure occurred.

4. 2. 3. Barrier option

Now we consider the new payoff :

$$\tilde{f}((X_t)_t) = (X_T - K)_+ \mathbb{1}_{\{\min_{s \leq t} X_s \geq L\}} = f((X_t)_t) \cdot \mathbb{1}_{\{\min_{s \leq t} X_s \geq L\}}$$

where L is a give treshol and f is the previous payoff (European call). This payoff is called the **barrier option**. The term $\mathbb{1}_{\{\min_{s \leq t} X_s \geq L\}}$ needs a specific treatment to be computed. To do so, we take interest into the function q given by :

$$q(t, s, a, b) = \mathbb{Q}(\exists r \in [t, s] \text{ such that } X_r < L \mid X_t = a, X_s = b)$$

The link between q and $\mathbb{1}_{\{\min_{s \leq t} X_s \geq L\}}$ comes from the fact that q verifies :

$$q(t, s, a, b) = \mathbb{Q}\left(\min_{r \in [t, s]} X_r < L \mid X_t = a, X_s = b\right)$$

The goal of the 6 following steps is to compute q . The 7th focuses on code.

1. The reflection principle :

Let us consider a standard Brownian motion W on $[0, t]$. By the **reflection principle**, we have, for any $m \geq w$:

$$\mathbb{P}\left(\max_{s \leq t} W_s \geq m, W_t \leq w\right) = \mathbb{P}(W_t \geq 2m - w) \quad (*)$$

Here is a graph to understand the principle :

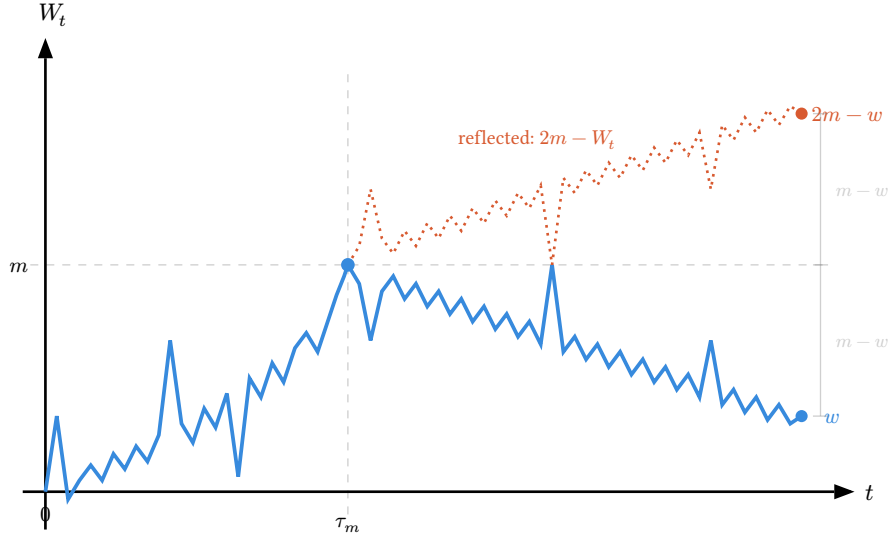


Figure 3: Reflection principle: the path after τ_m and its mirror image about $y = m$.

This principle allows us, thanks to the fact that W and $2m - W$ follow the same law on $[\tau_m, t]$, to get a new expression for the probability of the maximum of a Brownian motion intersected with an end-point condition.

2. Forcing the minimum to appear

We are interested in the $\min_{s \leq t} X_s$ more than in its maximum, but as :

$$\begin{aligned} \mathbb{P}\left(\min_{s \leq t} W_s \leq m, W_t \geq w\right) &= \mathbb{P}\left(-\max_{s \leq t} (-W_s) \leq m, -W_t \leq -w\right) \\ &= \mathbb{P}\left(\max_{s \leq t} (-W_s) \geq -m, -W_t \leq -w\right) \end{aligned}$$

We fall within the scope of the *reflection theorem*, since $-W$ is also a standard Brownian motion.

3. Conditional probability

We need to convert this intersection probability into a conditional one, to make a Brownian bridge appear. Thus :

$$\mathbb{P}\left(\min_{s \leq t} W_s \leq m \mid W_t \geq w\right) = \frac{\mathbb{P}\left(\min_{s \leq t} W_s \leq m, W_t \geq w\right)}{\mathbb{P}(W_t \geq w)}$$

4. Converting the inequality conditions into equality ones

We do not really know how to deal with these “ $| W_t \geq w$ ”, what we do know is the *Brownian bridge*, that needs equality conditions instead of inequalities. Let's then take $\varepsilon > 0$. The idea is that we consider inequality conditions on very short intervals around w that shrink as $\varepsilon \rightarrow 0$:

$$\begin{aligned}
\mathbb{P}\left(\min_{s \leq t} W_s \leq m \mid W_t \in [w, w + \varepsilon]\right) &= \mathbb{P}\left(-\max_{s \leq t}(-W_s) \leq m \mid -W_t \in [-w - \varepsilon, -w]\right) \\
&= \mathbb{P}\left(\max_{s \leq t}(-W_s) \geq -m \mid -W_t \in [-w - \varepsilon, -w]\right) \\
&= \frac{\mathbb{P}\left(\max_{s \leq t}(-W_s) \geq -m, -W_t \in [-w - \varepsilon, -w]\right)}{\mathbb{P}(-W_t \in [-w - \varepsilon, -w])} \\
&= \frac{G_t(-w) - G_t(-w - \varepsilon)}{\Phi_{-W_t}(-w) - \Phi_{-W_t}(-w - \varepsilon)} \\
&\text{where } \begin{cases} G_t(x) = \mathbb{P}\left(\max_{s \leq t}(-W_s) \geq -m, -W_t \leq x\right) \\ \Phi_{-W_t}(x) = \mathbb{P}(-W_t \leq x) = \Phi_{W_t}(\text{symmetric}) \end{cases} \\
&= \frac{G_t(-w) - G_t(-w - \varepsilon)}{\varepsilon} \cdot \frac{\varepsilon}{\Phi_{W_t}(w + \varepsilon) - \Phi_{W_t}(w)}
\end{aligned}$$

Now, by $\varepsilon \rightarrow 0$ and (*), we deduce :

$$\mathbb{P}\left(\min_{s \leq t} W_s \leq m \mid W_t = w\right) = \frac{f_t(2m - w)}{f_t(w)}$$

by the reflection theorem, where f_t is the density of $\mathcal{N}(0, t)$.

5. Computing the wanted probability

At last, as $f_t(2m - w)/f_t(w) = \exp(-\frac{2}{t}m(m - w))$, we can compute the probability we've been looking for :

$$\mathbb{P}\left(\min_{s \leq t} W_s \leq m \mid W_t = w\right) = e^{-\frac{2}{t}m(m-w)} \quad (**)$$

6. Linking back to the initial problem

We now have the $(**)$ formula for any standard Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and we want

$$q(t, s, a, b) = \mathbb{Q}\left(\min_{r \in [t, s]} X_r < L \mid X_t = a, X_s = b\right)$$

We know that under $\mathbb{Q}$, Y is a Brownian motion starting at $g(x)$. Let's change our two formulae to express them in terms of Y and $\mathbb{Q}$ instead of X and $\mathbb{P}$:

$$q(t, s, a, b) = \mathbb{Q}\left(\min_{r \in [t, s]} Y_r \leq g(L) \mid Y_t = g(a), Y_s = g(b)\right)$$

which is true because g is strictly increasing. Now, $(W_r)_{r \in [0, t]}$ in the formula $(**)$ has to be a standard Brownian motion while $(Y_r)_{r \in [t, s]}$ is not, as it has a non necessarily zero starting point $g(a)$. Hence we shift the time and define $(\tilde{W}_r)_{r \in [0, s-t]} := (Y_{r+t} - g(a))_{r \in [0, s-t]}$ which is a standard Brownian motion on $[0, s-t]$.

This leads us to consider :

$$\begin{aligned} & \mathbb{Q}\left(\min_{r \leq s-t} Y_{r+t} - g(a) \leq m \mid Y_{s-t+t} = w + g(a)\right) \\ = & \mathbb{Q}\left(\min_{r \in [t, s]} Y_r \leq m + g(a) \mid Y_s = w + g(a)\right) \end{aligned}$$

At last, we need $\begin{cases} g(a) + m = g(L) \\ w + g(a) = g(b) \end{cases} \implies \begin{cases} m = g(L) - g(a) \\ w = g(b) - g(a) \end{cases}$, that we plug into our formulae to get :

$$q(t, s, a, b) = \exp\left(-\frac{2}{s-t}(g(L) - g(a))(g(L) - g(b))\right)$$

7. Code

Now we know the exact value of

$$q(t, s, a, b) = 1 - \mathbb{Q}\left(\min_{r \in [t, s]} Y_r \geq g(L) \mid Y_t = g(a), Y_s = g(b)\right) \quad \forall t, s$$

Which implies that, given the $P \sim \mathcal{P}(TB)$ intermediate points (τ_i, V_i) that we compute for *Beskos-Roberts* :

$$\begin{aligned}\mathbb{Q}\left(\min_{[0,T]} Y_r \geq g(L) \mid Y_T\right) &= \prod_{i=1}^{P+1} \mathbb{Q}\left(\min_{[\tau_{i-1}, \tau_i]} Y_r \geq g(L) \mid Y_{\tau_{i-1}}, Y_{\tau_i}\right) \\ &= \prod_{i=1}^{P+1} \left[1 - q\left(\tau_{i-1}, \tau_i, g^{-1}(Y_{\tau_{i-1}}), g^{-1}(Y_{\tau_i})\right)\right] \\ &= \prod_{i=1}^{P+1} \left[1 - \exp\left(-\frac{2}{\tau_i - \tau_{i-1}} (g(L) - Y_{\tau_{i-1}})(g(L) - Y_{\tau_i})\right)\right]\end{aligned}$$

where we took $\tau_0 = 0, \tau_{P+1} = T$ by convention.

Thus :

$$\begin{aligned}\mathbb{E}^{\mathbb{P}}[\tilde{f}(X)] &= \mathbb{E}^{\mathbb{Q}}\left[\tilde{f}(g^{-1}(Y)) \cdot e^{h(Y_T) - h(Y_0) + \int_0^T c(Y_s) ds}\right] \\ &= \mathbb{E}^{\mathbb{Q}}\left[f(g^{-1}(Y)) \cdot \mathbb{1}_{\{\min_{[0,T]} Y_s \geq g(L)\}} \cdot e^{h(Y_T) - h(Y_0) + \int_0^T c(Y_s) ds}\right]\end{aligned}$$

That we can compute by Monte-Carlo by the following method :

- Doing the same thing as in Beskos-Roberts method on European call for $f(g^{-1}(Y))$, $e^{h(Y_T) - h(Y_0)}$, $e^{\int_0^T c(Y_s) ds}$

- $\mathbb{1}_{\{\min_{[0,T]} Y_s \geq g(L)\}}$: we can compute

$$\prod_{i=1}^{P+1} \left[1 - \exp\left(-\frac{2}{\tau_i - \tau_{i-1}} (g(L) - Y_{\tau_{i-1}})(g(L) - Y_{\tau_i})\right)\right]$$

with the (τ_i, V_i) we have already simulated to get $e^{\int_0^T c(Y_s) ds}$.

Note : for the code, as soon as a τ_i verifies $Y_{\tau_i} \leq g(L)$, we set the indicator function to 0 without further computations.

4. 2. 4. Tests for the barrier call

1. European > Barrier

The first test we ran was just meant to check a property that would have allowed us to instantly detect an issue with the method if violated. That is the fact that, as $M \rightarrow \infty$, we expect the price of the *European call* to be greater than the price of the *barrier call*. Of course, for a few

paths simulated (low M), we could observe the opposite. But for $M = 10^6$, we start to get that property regardless of the other parameters. Thus, we **pass** this test.

2. Price decreasing in L and converging to the European as $L \rightarrow 0$

The second test we ran consists of the next plot :

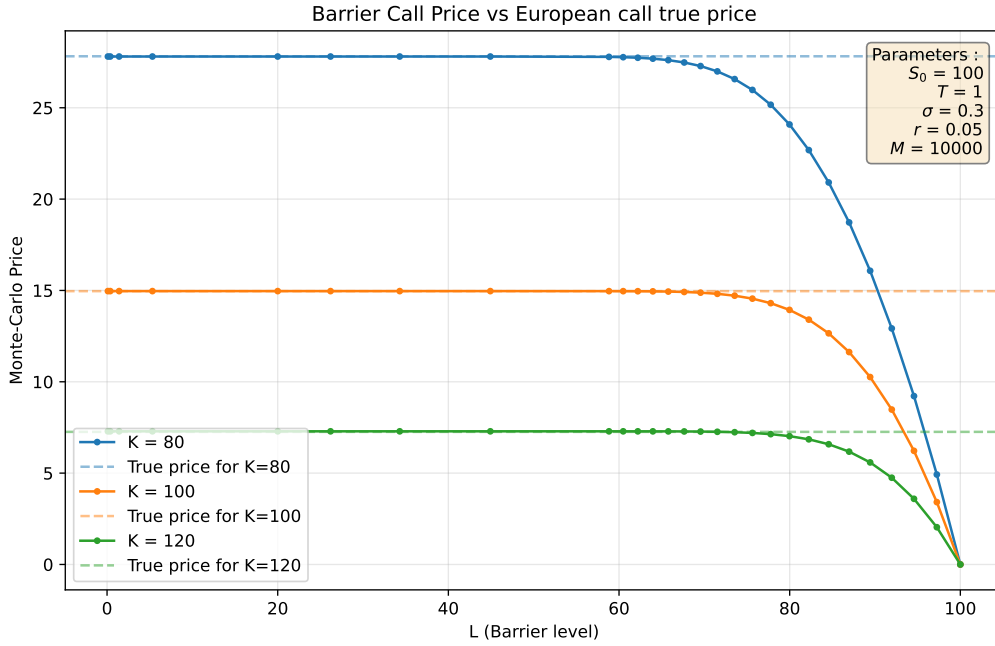


Figure 4: Barrier call convergence and monotony

Here is what we can say about it :

- There is a plateau at the European price for low L values. This is the convergence we were expecting : the lower the barrier, the more (X_t) paths reach $t = T$ without hitting it. This is exactly the European payoff behavior, hence the proximity. For $K = 80$, the plateau is reached as soon as $L \simeq 60$, but for $K = 120$, it is reached at $L \simeq 80$. This is because for low K , the payoffs live closer to the barrier, so more of them hit it and disconnect from the plateau. But for high K , the only surviving payoffs are the higher ones, which evolved far above the barrier, having less chance of hitting it.
- The previous point has for consequence that you can approximate the problem of pricing a barrier call by pricing the equivalent European call, for low L values. This “low” criterion **depends on σ and T** : the higher the volatility, the higher the probability that the path

hits the barrier and disconnect from the European and the longer T the more opportunities (X_t) has to fall under the barrier too.

- The price is decreasing in L as expected. This is because the higher L the more paths hit it, and the lower L the bigger the surviving area. For L close to S_0 , we expect almost all paths (that start at S_0) to hit the barrier, ending up in a payoff of 0.
- This test is qualitative : it gives us a trend that we compare to what we expected, but it doesn't allow us to assert our method works nor that the code is correct.

3. Quantitative test

This last test asserts that this $q(t, s, a, b)$ computing method, combined with Beskos-Roberts, is actually an unbiased method that computes $\mathbb{E}^\mathbb{P}[f((X_t)_t)]$ for f the barrier payoff. To do so, we adapted the Euler scheme we first used for the European call to the barrier call. We expect it to be a biased estimator of the barrier call price that overestimates it. The reason is that Euler scheme discretizes $[0, T]$ into a grid $\mathcal{G} = \{0 + i\frac{T}{N}, i \in \llbracket 0, N \rrbracket\}$ that might miss a point $\tau \notin \mathcal{G}$ such that $X_\tau < L$. The discretized path $\tilde{X}^N$ might verify $\min_{t \leq T} \tilde{X}_t^N \geq L$ while the actual path X has $\min_{t \leq T} X < L$. Hence the overestimation :

$$\mathbb{E}^\mathbb{P} \left[f \left((\tilde{X}_{t_i}^N)_{t_i \in \mathcal{G}} \right) \right] \geq \mathbb{E}^\mathbb{P} [f((X_t)_t)]$$

As an estimator, we also expect from $\tilde{X}$ that :

$$\mathbb{E}^\mathbb{P} [f((\tilde{X}^N))] \xrightarrow{N \rightarrow \infty} \mathbb{E}^\mathbb{P} [f(X)]$$

Finally, the last thing our test has to show is the $O(1/\sqrt{N})$ convergence rate of the Euler scheme. Here is the result of the test, where we draw the curves in $\frac{1}{\sqrt{N}}$ instead of N to identify the convergence rate instantly :

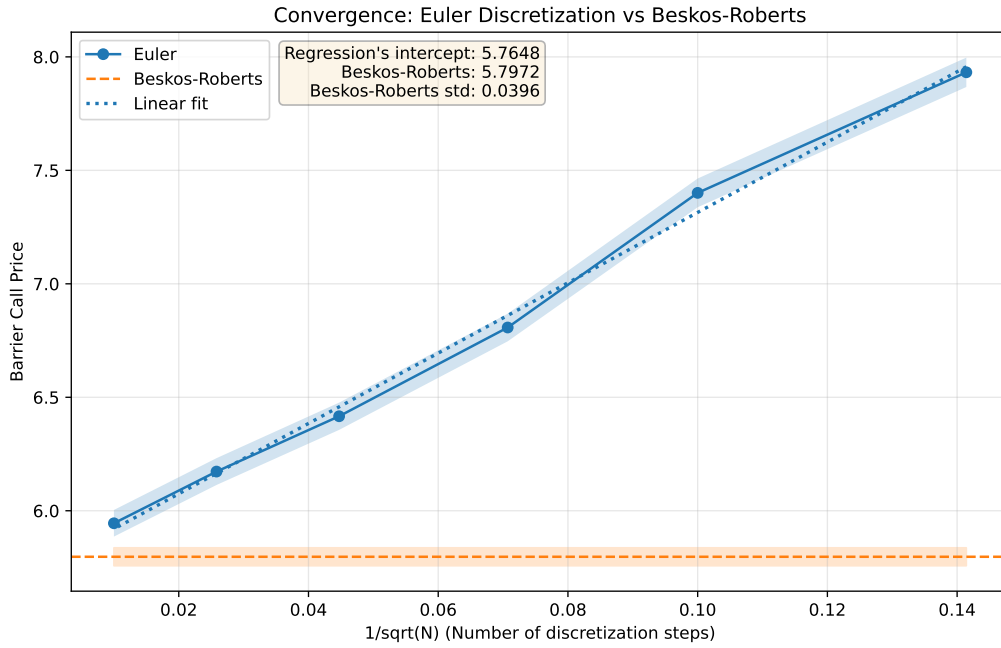


Figure 5: Euler scheme converging to barrier call unbiased method.

As expected, the Euler scheme overestimates the true price as it is decreasing. The linear regression in $1/\sqrt{N}$ shows that the Euler scheme converges :

- with a $O(1/\sqrt{N})$ rate
- toward the Beskos-Roberts price : intercept $\in [\mu_{BR} \pm \sigma_{BR}]$

Thus, Beskos-Roberts is an unbiased estimator of the barrier call price.

The method is **approved** by these 3 tests.